Marten Weijer

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About

I am a master's student in Data Science at Minnesota State University, Mankato, advised by Prof. Rushit Dave. I'm drawn to questions about how AI is changing software development, and I want to use my PhD to find where I can contribute most. My interests include AI for software engineering, empirical software engineering, and software analytics. I am applying to PhD programs for Fall 2027.

Research

The Missing Middle: A Systematic Review of Machine Learning for Municipal-Level Residential Energy Forecasting
M. Weijer, R. Dave, M. Bhavsar
2026 IEEE International Conference on Emerging Trends in Networks and Computer Communications (ETNCC), Windhoek, Namibia, 2026
Accepted and presented; proceedings forthcoming

Interpretable Machine Learning for Municipal Residential Gas Consumption: Integrating Climate, Building Stock, and Socioeconomic Drivers
M. Weijer, R. Dave, M. Bhavsar
Sustainability (MDPI), Special Issue on Artificial Intelligence, Image Processing, and Data-Driven Solutions for Sustainable Development, 2026
Accepted

Education

Minnesota State University — Mankato, MN
Master of Science in Data Science — *Jan. 2026 – May 2027*

Amsterdam University — Amsterdam, The Netherlands
Bachelor of Science in Computer Science — *Aug. 2013 – May 2017*

Experience

Graduate Teaching Assistant — Aug. 2026 – Present
Minnesota State University – CIS 483: Web Applications and User Interface Design — *Mankato, MN*

- Grade assignments, hold lab office hours, and give code-review feedback on student web applications

Senior Software Engineer — Feb. 2021 – Jan. 2026
Booking.com — *Amsterdam, The Netherlands*

- Designed and maintained large-scale data systems, with a focus on testing, code review, CI and reliable delivery across teams
- Built and operated backend services and event-driven pipelines in Java, Spring and Kafka, deployed on Kubernetes
- Led design discussions and code reviews, and mentored engineers on testing practices and maintainable code
- Took part in on-call rotations, debugging production incidents and following up with root-cause analysis

Software Engineer — May 2017 – Feb. 2021
Capgemini B.V. — *Amsterdam, The Netherlands*

- Developed Python and SQL pipelines and automated test suites, working in version-controlled, CI-driven development workflows
- Built and maintained backend applications and REST APIs for enterprise clients across several projects
- Worked in Scrum teams, turning client requirements into technical designs and shipping features in regular sprints

Certifications

Kafka: Confluent Certified Developer for Apache Kafka (CCDAK) — 2024
SAFe: Certified SAFe Practitioner — 2024
Spring: Spring Certified Professional for Java Developers — 2024
Kubernetes: Certified Kubernetes Application Developer (CKAD) — 2023
Scrum: Professional Scrum Master I — 2023
Java: Oracle Certified Java SE Developer — 2022